Facial Emotion Recognition using Deep Learning

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Abstract - **As it is known, sentiments influence information processing, attitude formation, and decision making to a great extent in real-world scenarios. Several recent efforts have been published about FER or facial expression recognition, however, due to the diversity of human faces and fluctuations in pictures, reliable and robust FER systems remain a challenge. Till date, every study and work has proposed either a single network or an ensemble model. The accuracy of ensemble models is higher, but they were associated with many models and datasets and a few tweaked datasets to improve the accuracy, increasing the computing complexity. While the majority of research in this field focuses on improving accuracy, this study utilizes the proposed model to a real-world scenario in which a person's face contains a mix of emotions, and a single-label sentiment can be highly noisy in such situations. In view of this scenario, we developed and tested 15-20 models and methods. In this paper, we propose a single standalone-based CNN model with its implementation on a real-time Intelligent System for Sentiment Recognition, which validates accuracy through transfer learning and performs tasks such as face detection, sentiment classification, and providing a live list of probabilistic labels in Realtime from a webcam feed in one blended step. The proposed model achieves an accuracy of 76.62 % outperforming all standalone-based models like VGG16, VGG19, EfficientNetB7, and other proposed models on the FER2013 dataset solely, which is a challenging and noisy crowd-sourced dataset, without using any other auxiliary dataset.**

Keywords: Deep learning, Facial emotion, Recognition, Human sentiment

I. Introduction

According to the research by psychologist Mehrabian on human information communication, 55 percent of information is communicated by facial sentiments, 38 percent via supporting languages such as sound, voice, and just 7 percent through spoken language [1]. Despite deep learning approaches showing efficiency of automatic facial sentiment recognition, the study of emotions had started gaining interest in the early 1970s. Ekman and Friesen introduced six basic emotions: anger, disgust, fear, happiness, sadness, and surprise [2]. Since the late 1980s, computer science academics have been interested in emotion research, culminating in published studies on how to develop automated emotion recognition systems. Most of the traditional approaches consist of three steps, the first step in practice is to detect the position of the face, followed by extraction of facial geometric features to generate specific vectors and then classifying the sentiments with maximum score [3]. A lot of technical manipulation is required in these methods. The characteristic analysis becomes extremely challenging as the data is massive. CNN started showing great potential as soon as it was introduced, in the late 1990s. It has proved itself in a wide variety of image classification tasks [4], and it has also been applied in emotion recognition. One of the primary appeals of Deep Learning CNNs is its ability to learn features directly from data, avoiding the tedious and monotonous hand-crafted feature generation, used in other supervised learning strategies. However, the use of CNNs was limited at that time, because of lack of training data and processing capacity. As the processing power and the accumulation of bigger datasets improved after the 2010s, CNNs became a far more feasible tool in image processing, pattern identification, and feature extraction. In this paper, the face is detected by haar cascade feature extraction and sentiment classification is done by different layers of the CNN model. A Facial Sentiment/Emotion Recognition framework can be a significant piece of computerized reasoning and can fill in, certifiable applications in various fields like mental investigation, observing driver's exhaustion, intelligent game plan, versatile applications to naturally embed feelings in speech, helping frameworks for medically introverted individuals, a robot with enthusiastic insight, etc. The quality and quantity of the training dataset largely determine the overall performance of the final system. From the confusion matrices, generated by models developed at the early stages of this work, it was observed that sentiments like disgust and fear have an accurate prediction rate of less than 60%. Considering the real-world scenarios and the importance of particular emotions in such systems, we reduced the feature extraction to five sentiments, and the final proposed model is trained and tested for sentiments like happy, sad, angry, surprise, and neutral. As it is observed that disgust and fear are noisy even for humans and a great portion of the labelling done for these two sentiments is mislabelled proving their noisy nature and is hindering the accuracy of the model to 71.02%. On further studying, we realized that there are scenarios where a person has mixed emotions and for such scenarios single labelled emotions are very noisy and there can be a need of predicting the next dominant facial sentiment present. For example, Trohidis et al. noticed that music may elicit many emotions at once and evaluated four multilabel classification systems to solve the problem [5].

II. Literature Review

A. Studies made

Many CNN variants have achieved remarkable results with a classification accuracy between 65 % and 76.82 % [6]. Assembling multiple models has shown improved performance. For instance, in order to enhance performance, Liu et. al. trained three distinct CNNs and combined them. The greatest single-network accuracy they obtained was 62.44 percent [7]. Minaee et. al. obtained a 70.02 percent accuracy using an attentional convolutional network in an end-to-end deep learning framework [8]. Tang et. al. replaced the softmax

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layer with a support vector machine in a deep neural network and achieved a classification accuracy of 71.52 % [9]. Shi et. al. suggested a unique amended representation module (ARM) to replace the pooling layer and obtained 71.38 percent testing accuracy [10]. Pramerdorfer et. al. compared the performance of VGG architecture, Inception architecture, and ResNet architecture. According to their findings, VGG had the greatest accuracy of 72.7 percent, followed by ResNet at 72.4 percent and Inception at 71.6 percent. Other studies have attempted to enhance their model's performance on FER2013 by improving the data and incorporating supplementary training data, however, this is outside the scope of this study.

B. Datasets and their reported results

A competition named Facial Expression Recognition Challenge was held in 2013. During this competition, competitors were encouraged to build the best system for detecting which emotion is being conveyed by a picture of a human face in a facial expression identification competition. Pierre Luc Carrier and Aaron Courville created and launched the 'Facial Expression Recognition 2013' (FER-2013) dataset in this competition. The dataset was produced by using the Google image search API. There are 4953 "Anger" photos, 547 "Disgust" images, 5121 "Fear" images, 8989 "Happiness" images, 6077 "Sad" images, 4002 "Surprise" images, and 6198 "Neutral" images among the 35887 images. Ian Goodfellow conducted several small-scale tests to assess human performance on this job and discovered that human accuracy on the FER-2013 was 65+5%. The top four teams utilized discriminatively trained convolutional neural networks with image transformation, as illustrated in Table No. 1. [11] Each work done in this field after the year 2015, using FER2013 dataset has been mentioned in Table No. 2 along with their reported accuracies.

TABLE I. TOP 4 ACCURACIES OBTAINED DURING THE CHALLENGE[11]

Team	Members	Accuracy
RBM	Yichuan Tang	71.162%
Unsupervised	Yingbo Zhou, Chetan Ramaiah	69.267%
Maxim Milakov	Maxim Milakov	68.821%
Radu + Marius + Cristi	Radu Ionescu, Marius Popescu, Cristian Grozea	67.484%

Table 2. Work and Researches done using the FER2013 dataset

S. No	Method	Dataset	Accuracy	Author/year
1	sequential fully-CNN Less no of parameters trained with ADAM optimizer	FER2013 IMDB	66%	Octavio Arriaga et al [2018] [12]
2	Image based static facial expression recognition with multiple deep network learning	FER2013	72.00%	Yu et al [2015] [13]
3	MRMR-based ensemble pruning for facial expression recognition.	FER2013	70.66%	Li et al. [2017] [14]
4	Using a Hybrid CNN–SIFT Aggregator.	FER2013	73.4%	Connie et al. [2017] [15]
5	Human Emotions Recognition for Realizing Intelligent Internet of Things	FER2013	71.91%	Hua et al. [2019] [16]
6	Local Learning with Deep and Handcrafted Features	FER2013	73.25%	Mariana-Iuliana Georgescu [2020] [17]
7	CNN OpenCV implemented in the Raspberry Pi consists of 3 main processes detection, feature extraction, emotion classification.	FER2013	65.97%	Lutfiah Zahara Purnawarman Musa [2020] [18]
8	Saptial features VGG CNN capture spatial and audio features from videos.	FER2013 EmotiW	60.03 %	Boris Knyazev Roman Shvetsov [2017] [19]
9	adopt the VGGNet architecture, fine-tune its hyperparameters, various optimization methods, saliency maps	FER2013	73.28 %	Yousif Khaireddin Zhuofa Chen [2021] [20]
10	SVM, saliency maps extract important regions for detecting different facial expressions	FER2013,CK +, FERG, JAFFE.	70.02%	Shervin Minaee, Amirali Abdolrashidi [2019] [21]
11	CNN, Random Search algorithm applied on a search space defined by discrete values of hyperparameters	FER2013	72.16%.	Adrian Vulpe-Grigoraşi Ovidiu Grigore [2021] [22]

III. METHODOLOGY

The final model was trained on sets excluding the sentiment of fear and disgust. Hence, the dataset was reduced to a total of 24282 images in the train set, 5937 images in the validation set, and 6043 images in the test set.

A. Data Augmentation

The first thing we did in our model is to account for the variability in facial expression recognition. We applied a significant amount of data augmentation in training images using Keras image data pre-processing and image data generator. This augmentation includes rescaling of the image data, also referred as normalizing, in the range of 0 to 1, which is preferred for neural network models, by dividing each pixel by 255. Rotation of image is done by 30 degrees, shearing and zooming of the image is done by a factor of 0.3, horizontal and vertical shifting of the image is done by a factor of 0.4 and finally, the image is flipped horizontally with the fill mode set to "nearest". Each of the technique is applied randomly. For validation and testing, images are only normalized by dividing each pixel by 255. After this, the image is set to grayscale and cropped to a size of 48×48 pixels, in the generator. As the shuffle is set to true, 20 random images at a time are sent from the train folder directory in a 'categorical' class mode which means that the labels are encoded as a categorical vector. And the same is followed for the validation generator.

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B. Model Description

The model we proposed is inspired from VGG-16 and is a sequential model, incremented via the add () method, which is provided by Keras since April 2020. The Sequential model is suited for a basic stack of layers with exactly one input tensor and one output tensor in each layer. Four convolutional stages and three fully linked layers make up the model. The convolutional stages are responsible for feature extraction, dimension reduction and nonlinearity whereas the fully connected layers are trained to classify the inputs as described by the extracted features. Each of the convolutional stages contains two convolutional blocks consisting of a convolutional layer, an ELU activation, and a batch normalization layer, with features increasing in each convolutional block as 32 in the first, 64 in the second, 128 in the third and 256 in the final block, following each block is a max-pooling layer and dropout layer. The above layers have been proposed to improve performance. For instance, the Rectified Linear Unit (ReLU) activation function has been replaced by ELU (Exponential Linear Unit) as activation to avoid gradient dispersion problems and speed up training. Max pooling is used to get max scores from the convolution block and down sample the inputs and aid in generalization. The Dropout layer is used consistently throughout to help prevent overfitting as a result train accuracy is kept in check and is lower than the test and validation accuracy. Batch normalization is used to speed up the learning process by preventing gradient vanishing and exploding.

The first fully connected layer is a dense layer in which data is passed after going through the flatten function and is then again followed by an ELU activation, batch normalization, and dropout layer. The second layer repeats the same layers without flattening the input and the third fully connected layer is a dense layer followed by 'SoftMax' activation as it is often used for the last layer of a classification network. The elements of the output vector are in the range (0, 1) and sum to 1.

Here, the flatten function converts or collapses the pooled feature map into the channel dimension which is a single column. After this the dense layer, which is an actual fully connected network layer, feeds the output of flattening function to all of its neurons and each of these neurons, generate a single output which is then again activated (through ELU) and normalized and then again feeds to the next two dense layers where later one gets activated by SoftMax function which predict a classification which in our case are five sentiments.

C. Learning rate and optimizer

One important aspect that might affect performance, among many others, is the learning rate. A low learning rate would considerably delay the model's convergence and might trap it in a nonoptimal local minimum. Oscillations around the minimum or loss divergence might result from a high learning rate. A learning rate scheduler, which adjusts the learning rate during training, is a useful approach. For example, as the number of iterations rises, time-based decay lowers the learning rate either gradually or dramatically. After a given number of epochs, the step decay reduces the learning rate by a factor. During training, an adaptive learning rate schedule attempts to automatically modify the learning rate based on local gradients. Cosine annealing, for example, regularly resets the learning rate and reuses "good weights" during the training process [20]. ReduceLROnPlateau is used to monitor test loss and lower the learning rate by 0.001, if no gain is detected after patience of up to 3 epochs.

The most commonly used optimizer is Stochastic gradient descent (SGD). It is a simple technique that updates the parameters of a model based on the gradient of a single data point. AdaGrad scales the learning rate for each network dimension adaptively. The learning rate is drastically reduced by RMSProp. Adam combines the benefits of AdaGrad and RMSProp by adjusting the learning rate and incorporating gradient momentum among other things, resulting in the best optimizer. In this model, the Adam optimizer with a learning rate of 0.001 was applied [20]. The test loss, which the model tries to reduce during training, is computed using loss functions.

D. Training

We ran all layers for 100 epochs optimizing the cross-entropy loss. The model is saved every time the maximum test accuracy is achieved. Early stopping is set with the patience of 10, i.e. to stop the training after 10 epochs if there is no improvement in validation loss and restore the model weights from the epoch with the best validated accuracy. Hence, the validation loss was also kept in check.

E. Material Used

To run the experiments, the machine used is a Lenovo Y510p laptop. GPU processing is not utilized in this project, only CPU power was utilized. The device specifications are as following:

- 4th Gen i7-4700MQ
- 12GB RAM
- 256GB SSD

The operative system used Windows 11 Pro 64bit.

Other software used includes Python 3.9.6 and TensorFlow 2.6.0. IDE, for implementation purposes like for execution of the code, training of the model, and interacting with the system developed is JetBrains's PyCharm 2021 for testing the application, DataSpell 2021 for testing different methods and CNNs and Jupyter notebook server 6.4.3 used as a web browser-based IDE for the final proposed model and plotting graph and confusion matrix using matplotlib.

IV. RESULTS

To properly analyse the proposed model, accuracy graph indicating train and test accuracy and loss graph indicating train and test loss are plotted using matplotlib against the number of epochs. As it is seen from Fig. 1 and Fig. 2, that in the first 20 epochs there is a sharp improvement in accuracy and loss. The model is able to outperform claimed human accuracy of 65% successfully at the 10th epoch and continue improving to a great extent, but after 40 epochs the loss and accuracy both tend to show very little progress and remain almost the same and around the 80th epoch stops improving further even at very low learning rates. The denial of rise in the train accuracy also shows that the model is not overfitted and the dropout layer has helped in achieving so.

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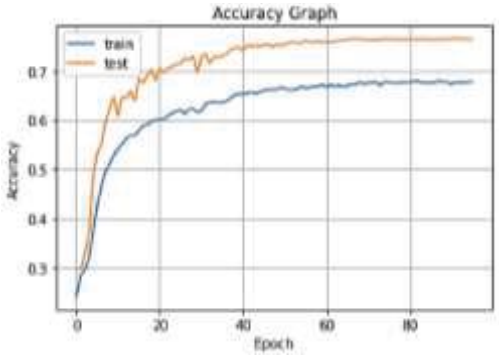


Fig. 1 Train and Test Accuracy

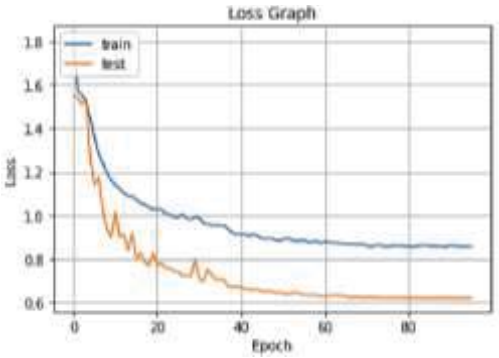


Fig. 2 Train and Test Loss

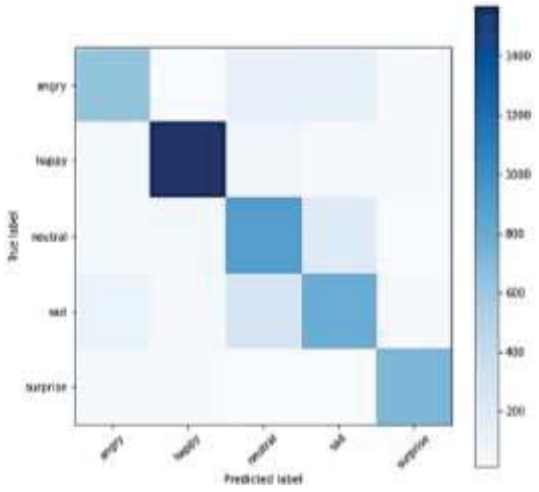


Fig. 3 Confusion Matrix

We can exhibit the confusion matrix together with the heatmap from Fig. 3 to better examine our model. Each row of the matrix represents a real setiment, while each column provides a projected sentiment. Table No. 3 shows the classification report for the testing data, which allows us to go further into the performance of each sentiment. As it can be seen, this system has the greatest precision in predicting 'happy' sentiments. and the lowest precision for 'sad'. The model's overall accuracy, macro avg, and weighted avg on a scale can also be seen all lying in the range of 0 to 1, where 1 represents 100%.

Table 3. Classification Report

	Precision	Recall	F1 score	Support
Angry	0.72	0.66	0.69	958
Happy	0.92	0.88	0.90	1774
Neutral	0.64	0.73	0.68	1233
Sad	0.66	0.64	0.65	1247
Surprise	0.86	0.87	0.87	831
Accuracy			0.76	6043
Macro Avg	0.76	0.76	0.76	6043
Weighted Avg	0.77	0.76	0.77	6043

V. APPLICATION

Coming to the implementation, the application is designed and developed from scratch, which not only shows the dominant sentiment of the face but also provides us with the list of probabilities of other sentiments which can be of great use in the real world scenario, where a person's faces carries two or more sentiments. Fig. 4 shows the structural flow of the application.

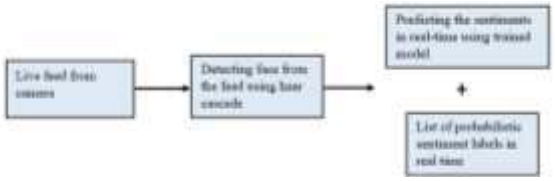


Fig. 4 The structural flow of the application

The face is first detected using the haar cascade as it is a robust face detection method. It reduces the computational cost and complexity that leads to high detection accuracy. After that, the image is sent to the trained model as an input and it returns the highest-scoring sentiment on the webcam feed in real time, along with the live list of probabilities for all other emotions in the canvas. Fig. 5 shows the result of the application developed.



Fig 5 Results of dominant sentiment on the webcam feed and their respective probability list

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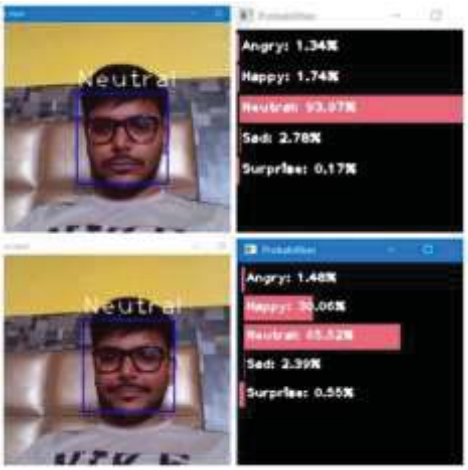


Fig 6 Prediction of mixed emotion.

Fig. 6 shows the real-world scenario where it can be seen that the dominant sentiment remains the same, that is, the neutral sentiment, but with the help of the live probability list, displayed adjacent to the webcam feed, it can be seen that happy is the second most dominant sentiment present which would have been ignored or missed if only 1 emotion would have been displayed by the system.

VI. Conclusion

To conclude, we closely observed and learned about various work and research done in this field. All implemented single standalone CNNs which are trained on the FER2013 dataset were studied and their achieved results were noted. We then developed, a single VGG inspired model and for that, the original model is closely studied and the contribution and importance of each layer in that model was observed. After this, we started developing the model from the scratch and after working on 10 to 15 models with different sets and combinations of layers, the final model presented in this paper was proposed, which achieved the highest accuracy of 76.62% , outperforming all single network models we know and studied in this work. Coming to the dataset, FER2013 remains a challenging dataset as it is wild and crowdsourced, the only modification done in the dataset was to drop two sentiments, which are fear and disgust as they are highly mislabelled and noisy. Furthermore, we designed and implemented a real-time Intelligent System for Sentiment Recognition, which accomplishes the tasks of face detection using haar cascade, sentiment classification using the proposed model's saved weights, and gives a live list of probabilistic labels simultaneously in real-time from a webcam feed in one blended step. In future we will implement our model to develop a web-based application using flask.

VII. Consent For Publication

We, Shubhanjay Pandey and Sonakshi Handoo, hereby confirm that the human images in Fig. No. 5 and 6 are ours and we give our consent for the publication of our identifiable details to be published.

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